

NEPA 3D Knowledge Graph Review System

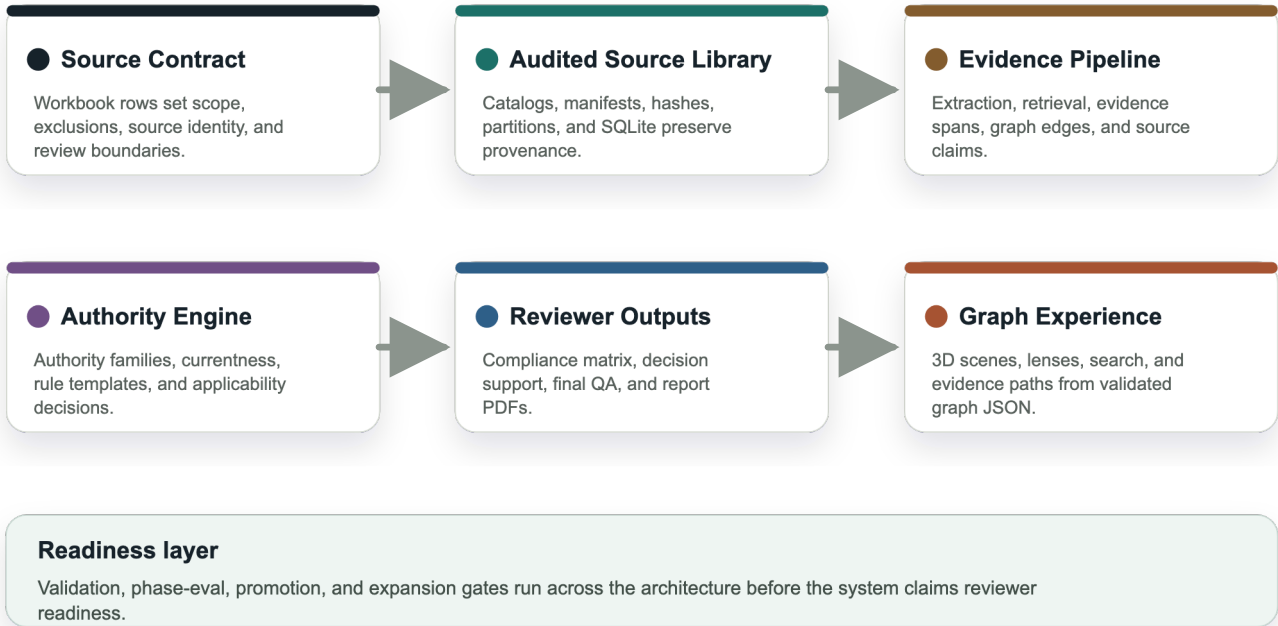
The system turns Region 1 NEPA source material into auditable review intelligence. It preserves source provenance, builds an authority graph, traces evidence to findings, and gates readiness before outputs reach reviewers. The result is a transparent operating layer for compliance review, decision support, and graph exploration.

Current Region 1 implementation snapshot.



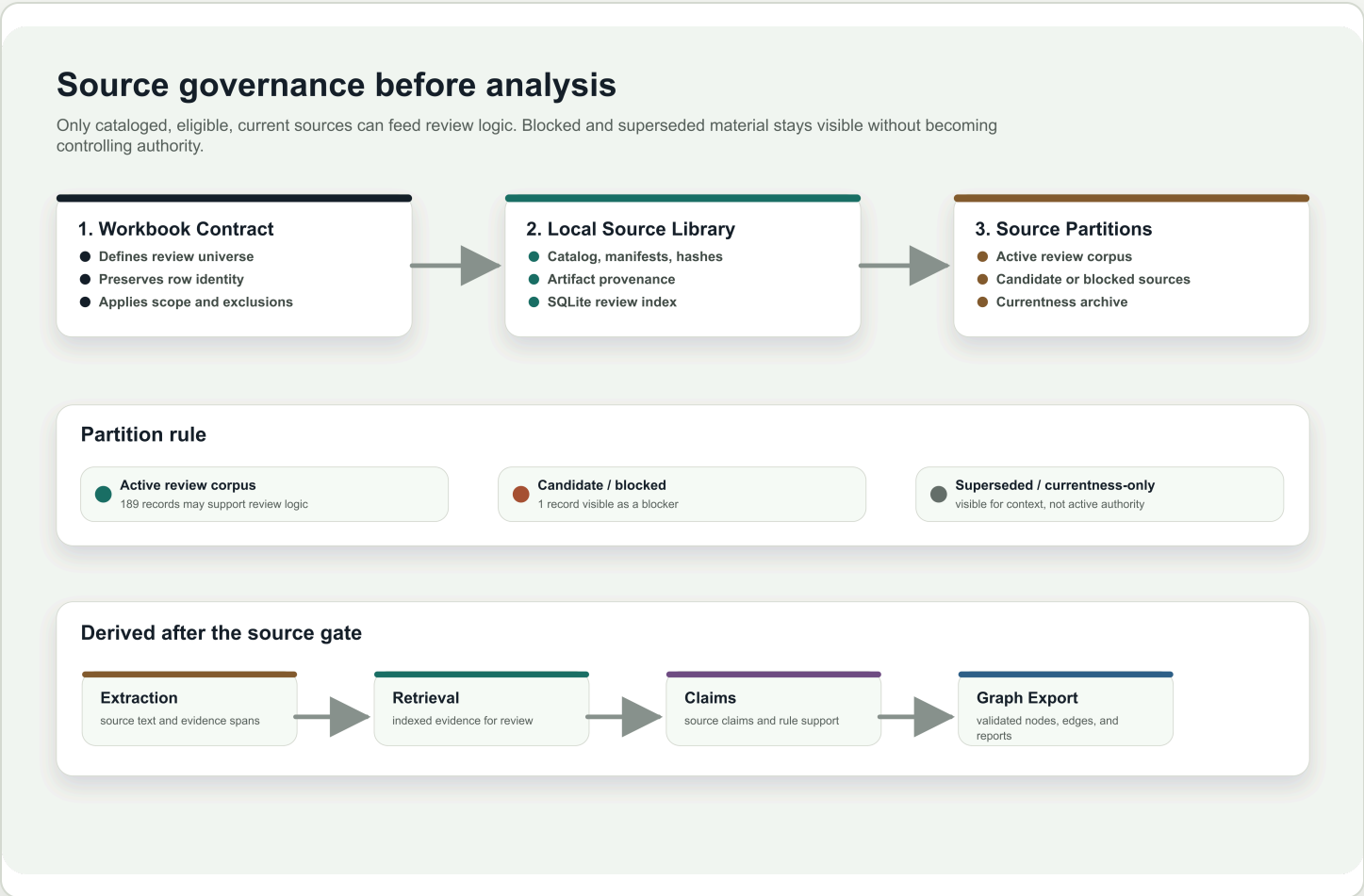
Architecture components at a glance

A source-governed review engine with traceable evidence, explicit authority logic, and gated outputs.



Controlled source library to graph-ready review layer

Every review starts with a controlled source contract. Workbook rows, exclusions, URL repairs, source partitions, catalog records, and artifact hashes define what can support review logic. Derived layers are rebuilt from those surfaces, not loose files.

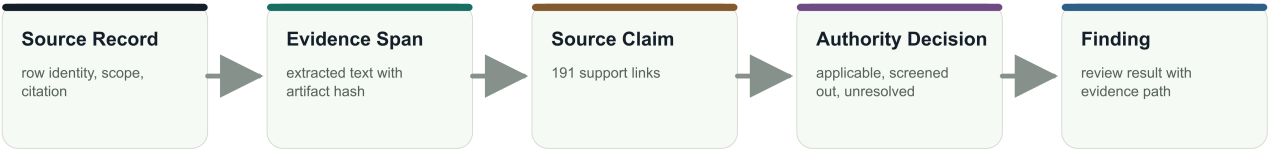


Every finding stays connected to evidence

Findings are produced from inspectable evidence paths. The system links source records, artifacts, evidence spans, claims, authority decisions, and review findings before it renders reports or graph views.

Finding trace

The review product is only the final view. The underlying path stays inspectable from source record to finding.



Reverse compliance uses the same path in the opposite direction.

Unsupported claims, stale authority language, missing evidence, and unresolved applicability are flagged before readiness.

Outputs generated from the trace



Traceability model: the finding is the end of a chain, not a free-standing conclusion. The same chain supports reports, QA replay, and graph exploration.

Audit surface

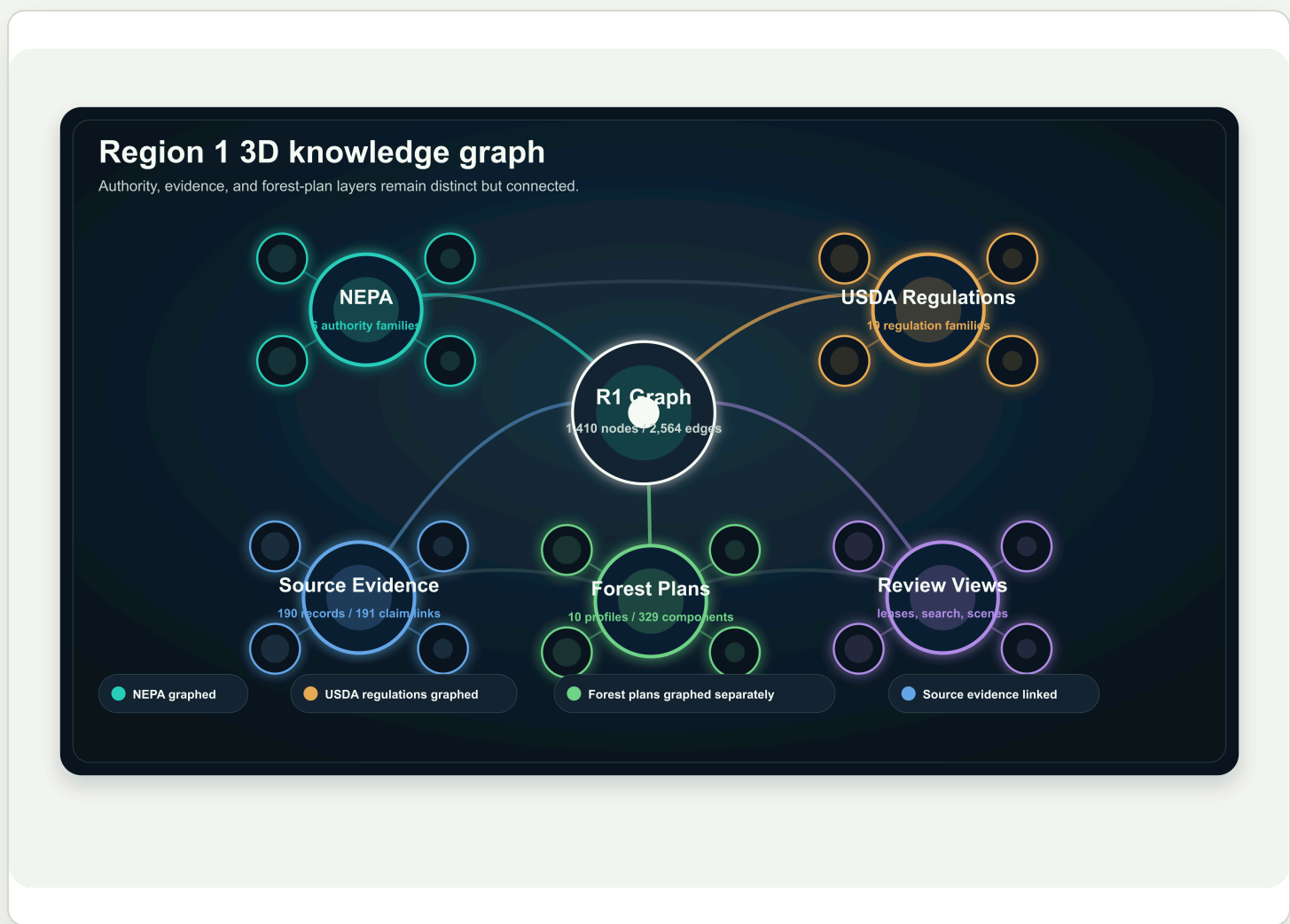
Each product can be traced back to source record, artifact hash, evidence span, and decision path. Reviewers can inspect the basis without relying on narrative alone.

Reverse compliance

The evidence model flags unsupported claims, older-regulation dependencies, missing source support, unresolved applicability, and forest-plan gaps before readiness is claimed.

NEPA, USDA regulations, and forest plans in one graph

The Region 1 graph is the client-facing surface for the system. It shows NEPA authority, USDA regulations and procedures, source evidence, and forest-plan layers as connected but distinct parts of the review universe.



Separate forest-plan layer

Forest-plan profiles and components are graphed separately from NEPA and USDA authority, then linked into review context when ready.

Authority layer

NEPA authority, USDA procedures, and regulatory families stay visible beside source evidence so reviewers can inspect how findings are supported.